

Week 04: Trees and strict evaluation

Functional Programming in Lean

Learning goals

- ▶ Recursion on an inductive tree
- ▶ Accumulator-based recursion
- ▶ Strict evaluation in Lean

Core concepts

- ▶ Tree cases: leaf and node
- ▶ Size from recursive subtree results
- ▶ Delayed computation: `Unit` $\rightarrow \alpha$

Lean example

```
def Tree.size : Tree  $\alpha$   $\rightarrow$  Nat
  | .leaf => 0
  | .node left _ right =>
    left.size + 1 + right.size
```

Lean details

- ▶ Persistent tree values
- ▶ Tail-recursive list reversal
- ▶ Termination from smaller subtrees

Live coding

- ▶ `Tree.size`
- ▶ `reverseAcc`
- ▶ Comparison with `List.reverse`

Check

Termination argument for `Tree.size`

Exercise

- ▶ `Tree.height`
- ▶ Tree-map size property